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EDUCATION

- **University of the Andes** Bogota, Colombia
M.Sc. in Systems and Computing Engineering – Graduate Assistant Aug 2026 – Present
Research Interests: Continual Learning, Deep Learning, Program Analysis, Compiler Optimization
- **National University of Colombia** Bogota, Colombia
Bachelor in Computer Science - GPA: 4.1/5.0 Aug 2020 – July 2025
Courses: Operating Systems, Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Analysis of Databases, Compilers, Cryptography, Coding Theory, Abstract Computational Algebra, Real Analysis, Numerical Systems

SKILLS SUMMARY

- **Languages:** Python, C++, Elixir, Julia, Ocaml
- **Frameworks:** PyTorch, Keras, Django, Hugging Face
- **Tools:** Docker, Azure, Weights Biases, MySQL, Elasticsearch
- **Platforms:** Databricks, Spark, Pinecone, GitHub CI/CD, GitLab CI
- **Soft Skills:** Technical Mentoring, Research Collaboration, Problem Solving, English Fluency

EXPERIENCE

- **DataLab, National University of Colombia** Bogotá, Colombia
Data Scientist – Modeling - Supervised by Dr. Fausto Moreno May 2026 – Present
 - **Risk Modeling:** Developed quantitative and geospatial risk models to characterize sanitary and productive risk factors for the Colombian livestock sector.
 - **Geospatial Data Science:** Integrated and analyzed municipal-level spatial datasets using Python, GeoPandas, and geographic information from national institutions.
 - **Data Integration:** Built reproducible pipelines to process and integrate heterogeneous datasets from organizations including ICA, UPRA, DANE, ADR, SNIES, and Porkcolombia.
 - **Spatial Indicators:** Designed risk indicators based on geographic accessibility, institutional presence, technical assistance, connectivity, and proximity to agricultural infrastructure.
 - **Decision-Support Analytics:** Produced analytical outputs, risk scores, maps, and visualizations to support territorial and sanitary decision-making.
- **Electrical Machines & Drives , National University of Colombia** Bogotá, Colombia
Research Assistant - Supervised by Prof. Javier Rosero Apr 2025 – Jul 2025
 - **Photovoltaic System Analysis:** Contributed to research on solar panel performance by analyzing electrical and environmental data to evaluate efficiency and reliability.
 - **Data Science:** Processed, cleaned, and analyzed large datasets from photovoltaic systems using Python (Pandas, NumPy, Matplotlib) to extract insights and support experimental conclusions.
 - **Modeling and Simulation:** Assisted in the development of predictive models for energy generation based on irradiance, temperature, and system characteristics.
 - **Collaboration:** Worked closely with faculty and graduate students to support experimental design, data acquisition, and result interpretation.
 - **Reporting:** Prepared documentation and visualizations for internal presentations and research reports.
- **Virtual Math Lab, National University of Colombia** Bogotá, Colombia
Research Assistant - Supervised by Prof. Juan Galvis Oct 2024 – Dec 2024
 - **Notebook Development:** Developed interactive Julia notebooks to support foundational mathematics courses such as Linear Algebra, Numerical Analysis, ODEs, Optimization, and Complex Variables.
 - **Educational Enhancement:** Created computational resources to help students consolidate understanding, explore practical examples, and improve problem-solving skills through hands-on execution.
 - **Open Educational Resources:** Published materials in the Labmatecc repository for free student access and local execution.
 - **Technical Support:** Maintained the lab's server and assisted students with technical questions regarding the use of Julia notebooks.
- **MindLab, National University of Colombia** Bogotá, Colombia
AI Lab Systems Administrator - Supervised by Prof. Fabio González Feb 2024 – Jul 2024
 - **Systems Administration:** Recovered inactive Linux servers after failures using advanced Linux/CUDA skills.
 - **Resource Optimization:** Configured CUDA cards for parallel processing acceleration of AI workloads.
 - **Technical Support:** Supported 15+ graduate students with lab resource access and utilization.
 - **Automation:** Implemented monitoring/automation solutions for AI research infrastructure.
 - **Documentation:** Developed and maintained server access guide for lab users.
- **Fregata Space** Remote
Machine Learning Intern Feb 2023 - Jul 2023
 - **Satellite Image Segmentation:** Developed DL models for land cover change detection and vegetation estimation
 - **Large-scale Processing:** Processed high-resolution satellite imagery using AWS infrastructure
 - **MLOps Implementation:** Automated model lifecycle management for segmentation pipelines
 - **Geospatial Analysis:** Extracted complex features from satellite data for environmental monitoring

PROJECTS

- **Energy-Aware Scientific Computing Research:**
 - Co-authored the research paper *The Growing Importance of Energy-Aware Numerical Precision in Scientific Computing* with Prof. Juan Galvis and Prof. Francisco Gómez
 - Studied the relationship between numerical precision, execution time, and energy consumption through experiments involving arithmetic operations, prime number generation, and LU factorization in Julia
 - Evaluated Int32, Int64, Float32, Float64, and mixed-precision computation using CPU energy measurements with RAPL
(2026 Jul)
- **Todos hacemos Ciencia – Scientific Communication Project:**
 - Founder and director of a science communication initiative focused on making scientific and academic topics accessible to broader audiences
 - Lead the planning, research, production, and dissemination of audiovisual content featuring scientific ideas, research, and academic experiences
 - Coordinate conversations and interviews with students, researchers, and academics to promote interest in science, research, and higher education
 - Manage the project's content strategy and YouTube presence, translating technical concepts into engaging and understandable discussions
(2026 Mar)
- **High-Performance Computing Laboratory Development:**
 - Collaborated with Prof. Juan Galvis and Prof. Francisco Gómez in the development and operation of a high-performance computing environment for students and researchers
 - Provided remote access to computing servers and technical guidance for scientific computing, numerical experiments, and computational research workflows
 - Assisted users with Linux environments, remote server access, software configuration, computational resource usage, and troubleshooting
 - Developed documentation and support resources to facilitate reproducible and efficient use of shared computing infrastructure
(2026 Feb)
- **DinoV2 Model Evaluation (Self-Supervised Learning):**
 - Compared prototype-based k-NN classification against full-data classification on ImageNet-A and ImageNet-R to evaluate representation quality under distribution shifts
 - Adapted the DINOv2 codebase to support efficient prototype construction and evaluation using learned visual embeddings
 - Analyzed the trade-off between classification performance, data usage, and computational cost when using class prototypes instead of the full training set
(Jan 2025)
- **Continual Learning Research (Computer Vision):**
 - Analyzed representation drift in ResNet models trained on CIFAR-10 under continual learning settings
 - Studied changes in learned feature representations and their impact on classifier performance across sequential learning stages
 - Evaluated strategies for mitigating classifier degradation, including Weight Aligning and prototype-based k-NN classification
(Jul 2024)
- **UNLOCK: Smart AI Agents – Navigating Monkeypox in Insurance (ML/NLP):**
 - Developed a scalable news analysis pipeline for retrieving and processing information related to monkeypox and insurance risk
 - Integrated Pinecone vector search and the Bing API to retrieve relevant and up-to-date information from heterogeneous news sources
 - Implemented an Amazon Bedrock-based intelligent agent to analyze retrieved information and support automated insurance risk assessment
(Jan 2024)
- **DATAPALOOZA: Data-Driven Decision Making (Data Engineering):**
 - Developed a data analytics pipeline and interactive dashboard for monitoring Amazon product, sales, and inventory information
 - Integrated external Amazon APIs to collect and process product data for downstream analysis and visualization
 - Built Power BI and Tableau visualizations to support data-driven analysis of sales performance and inventory management
(Jul 2023)

HONORS AND AWARDS

- **Bronze Award (4th place out of 111 teams) – Factored Datathon** Aug 2023
Two-week applied machine learning competition organized by Factored, with a focus on solving realistic business problems using NLP and modern AI tools. As the team's infrastructure and deployment lead, I was responsible for integrating and deploying Hugging Face models, designing a vector database pipeline using Pinecone, and ensuring fast and scalable document search. Our solution achieved high performance in both retrieval and reasoning tasks, placing 4th among 111 international teams.

VOLUNTEER EXPERIENCE

- **National University of Colombia** Bogotá, Colombia
Computer Science Mentor and Tutor 2024 – Present
 - **Student Mentorship:** Founded and lead the Comprehensive Support and Mutual Aid Group, a student mentorship initiative focused on providing sustained academic guidance, facilitating university integration, and encouraging students to take an active role in their own learning.

- **Long-Term Academic Support:** Mentored students at different stages of their undergraduate education through regular tutoring, Q&A sessions, academic guidance, and individualized support in computer science and mathematics.
- **Independent Learning:** Encouraged students to develop autonomy and intellectual curiosity by exploring topics beyond coursework, building personal projects, and developing self-directed learning habits.
- **Professional Development:** Connected students with hackathons, technology events, research activities, and external opportunities, helping them gain practical experience and broaden their academic and professional perspectives.
- **Community Building:** Built a peer-support community where students share knowledge, experiences, and opportunities, encouraging senior participants to mentor newer students and strengthen collaboration within the program.
- **Science and Technology Exposure:** Facilitated student participation in activities such as Google technology events and organized outreach initiatives, including an AI event in partnership with IBM.
- **Educational Resources:** Created and shared resources on academic development, university opportunities, and community initiatives, including documentation on student mentorship and mutual support.